

A Data Setup And Initial EDA Report

Chapter 1 : Introduction

1.1 Project Overview-

This project focuses on the initial analysis of two interconnected datasets: the Opportunity Dataset and the Learner Dataset. The primary objective of this phase is to establish a reliable database environment, import the datasets into PostgreSQL, perform exploratory SQL-based data profiling, assess overall data quality, and create initial visualizations that provide an understanding of the structure and characteristics of the data.

The Opportunity dataset contains information related to various opportunities such as internships, workshops, fellowships, scholarships, and other career development programs. It includes attributes such as opportunity category, duration, eligibility criteria, location, fees, rewards, application deadlines, descriptions, and administrative information.

The Learner dataset contains information regarding applicants who have applied to these opportunities. It includes application status, payment information, cohort allocation, learner responses, questionnaire data, completion status, withdrawal information, and other application lifecycle attributes.

Rather than directly performing data cleaning or business analysis, this phase concentrates on understanding the datasets from a technical and analytical perspective. A comprehensive data profiling exercise helps identify missing values, duplicate records, inconsistencies, and structural issues that may affect downstream analysis.

The outcomes of this phase establish a strong foundation for the subsequent stages of data cleaning, feature engineering, business intelligence reporting, and advanced analytics.

1.2 Project Objective-

1. Understand the real problem of the project.
2. Perform EDA and cleaning for making the data useful for analysis.
3. Find the relationships, patterns and trends follows in the both dataset.
4. Make the business decisions based on the analysis means which field have to focus, what is the strength and weakness area .

1.3 Project Understanding-

The project revolves around analyzing an ecosystem where organizations publish opportunities and learners apply for them. These two datasets together represent the complete lifecycle of an opportunity, from its creation to learner participation.

Opportunity Dataset

The Opportunity dataset stores detailed information about each opportunity published on the platform. Each record represents a unique opportunity and contains information such as:

Table 1.1 Opportunity Dataset Preview

Column	Meaning
pk	Primary Key. Unique identifier for each record in the database.
opportunity_id	Unique ID assigned to the opportunity. Often used for external references or APIs.
Badge	Special labels shown on the opportunity such as "Featured", "Trending", "Remote", "Top Employer", "New".
CareerAddOn	Additional career benefits like Certificate, Mentorship, Resume Review, Networking, Placement Assistance, etc.
category	Type of opportunity (Internship, Job, Competition, Scholarship, Workshop, Fellowship).
code	Internal code used by the company/system to identify the opportunity.
Cohort	Batch or intake number (e.g., Summer 2026, Batch 12, Cohort 5). Mostly used in training programs.
created_at	Date and time when the opportunity was created in the system.
currency_type	Currency used for fees or rewards (INR, USD, EUR, etc.).
current_editor	User/admin currently editing or responsible for maintaining the listing.
DropoutTransaction	Number of applicants who started but dropped out during the application process.
duration	Length of internship/course/program (e.g., 3, 6, 12).
duration_type	Unit of duration (Days, Weeks, Months, Years).
Eligibility	Eligibility criteria (e.g., B.Tech students, Final year, MBA, Freshers).
fee	Registration fee or application fee. Zero means free.
image_link	URL of the opportunity banner/logo/image.

Column	Meaning
is_archived	Boolean flag indicating whether the opportunity has been archived (1=True, 0=False).
is_auto_approve	Whether applications are automatically approved without manual review.
last_date_to_apply	Application deadline.
location	Internship/job location (Delhi, Remote, Bangalore, Hybrid, etc.).
long_description	Complete detailed description of the opportunity.
microscholarship	Small scholarship or financial incentive offered.
modified_at	Last time the opportunity was updated.
name	Title of the opportunity (e.g., Data Analyst Internship).
NotStartedTransaction	Number of selected candidates who have not yet started the program.
Panellist	Interviewer(s), evaluator(s), or judging panel assigned.
Reward	Prize, stipend, salary, certificate, cashback, etc.
role	Position being offered (Software Engineer Intern, Marketing Associate, etc.).
role_responsibility	Duties and responsibilities of the role.
short_description	Brief summary displayed in listings.
summary	Condensed overview of the opportunity.
Testimonial	Reviews or testimonials from previous participants.
tracking_questions	Custom questions applicants must answer while applying (e.g., "Why do you want this internship?").

Learner Dataset-

The Learner dataset records every learner application submitted for an opportunity. Each record represents a learner's application and captures the entire application journey.

1. Primary Keys & Relationships

Column	Meaning
pk	Primary key (unique learner/application record ID).
sk	Sort key or secondary key (used in databases like DynamoDB).
relation_id	Relationship ID linking learner and opportunity.
application_id	Unique application number.
work_item_sk	Internal workflow/process ID.
external_reference_url	External link related to the application.

2. Cohort Information

Column	Meaning
assigned_cohort	Batch assigned after selection.
cohort_p1	Phase 1 cohort.
cohort_p2	Phase 2 cohort.

3. Application Status

Column	Meaning
status	Current application status.

4. Important Dates

Column	Meaning
apply_date	Date learner applied.
created_at	Record creation date.

Column	Meaning
modified_at	Last update date.
accept_reject_date	Date of acceptance/rejection.
not_started_date	When learner failed to start.
completion_date	Program completion date.
reward_awarded_date	Reward issued date.
withdraw_date	Date learner withdrew.
dropped_out_date	Date learner dropped out.

5. Payment Information

Column	Meaning
fee	Registration/Application fee.
amount_to_be_paid	Remaining payment due.
transaction_id	Payment transaction number.
payment_status	Paid, Pending, Failed, Refunded, etc.

6. Team Information

Column	Meaning
team_name	Team name if team applications are allowed.
team_code	Team identifier.
team_creator	Person who created the team.

7. Acceptance / Withdrawal

Column	Meaning
accept_reject_reason	Reason for rejection or acceptance comments.
withdraw_reason	Why learner withdrew.

8. Learner Responses (Questionnaire)

These are responses collected during the application process.

Column	Meaning
why_do_you_want_to_apply_for_this_internship	Motivation statement.
why_do_you_want_to_apply_for_this_internship1	Another version of the motivation question.
can_you_share_an_experience_where_you_proactively_pursued_learning	Behavioral question.
please_identify_your_student_status	College student, Graduate, Working Professional, etc.
i_understand_that_i_will_have_to_commit_at_least56_hours_each_week	Agreement checkbox.
i_understand_that_this_is_the_first_step_towards_my_application	Consent confirmation.
i_will_be_available_from79_pm_ist730930_am_cst_for_meetings_on_c	Availability confirmation.
i_agree_to_terms_and_conditions_of_global_shala	Terms & Conditions acceptance.
i_have_read_and_agreed_to_the_terms_and_conditions	Another consent field.
send_for_approval_mechanism_t_c	Approval/consent confirmation.

Most of these are Yes/No fields.

9. Marketing Attribution

These tell where applicants discovered the opportunity.

Column	Meaning
from_where_did_you_hear_about_us	Referral source.
from_which_medium_did_you_hear_about_the_internship	Platform used.
how_did_you_hear_about_this_opportunity	Marketing channel.

Typical values

- Social media
- emailFromRochesterInstituteofTechnology
- emailFromSaintLouisUniversity
- myFriendRelative
- googleSearch
- Gwendolyn Brooks College Prep
- Gwendolyn Brooks College Prep
- Kruu Website, Saint Louis University

These are valuable for marketing analysis.

10. Reward Information

Column	Meaning
reward_awarded_date	Date reward/certificate was issued.

11. Wishlist

Column	Meaning
appear_in_wishlist	Indicates whether learner saved/bookmarked the opportunity before applying.

1.4 Relationship Between Both Datasets

The Opportunity dataset and Learner dataset are logically connected through the opportunity relationship. One opportunity can receive multiple learner applications, forming a one-to-many relationship.

This relationship enables future analyses such as:

- Applications per opportunity
- Opportunity popularity
- Completion rates
- Learner conversion funnel
- Cohort performance
- Marketing effectiveness
- Revenue generation
- Opportunity success analysis

Although this phase does not involve advanced business analytics, understanding this relationship is essential before proceeding to data transformation and dashboard development.

1.5 Tools and Technologies Used

The following technologies were used throughout this phase:

PostgreSQL

Used as the relational database management system to store, manage, and query both datasets efficiently.

Supabase

Used as the cloud-hosted PostgreSQL platform for importing datasets, managing database tables, executing SQL queries, and maintaining database connectivity.

SQL

Used extensively for:

- Data profiling
- Missing value analysis
- Duplicate detection
- Descriptive statistics

- Distribution analysis
- Data quality assessment

Looker Studio

Used for creating initial visualizations that summarize the structure and characteristics of the datasets.

CHAPTER 2

DATABASE SETUP

2.1 Database Creation

A PostgreSQL database was configured using the Supabase cloud platform. Separate tables were created for the Opportunity and Learner datasets to maintain proper data organization and enable efficient querying.

2.2 Data Import Process

The original CSV datasets were imported into PostgreSQL using Supabase's table import functionality. During the import process, the data types were automatically inferred and verified to ensure compatibility with the dataset schema.

2.3 Table Creation

Two primary tables were created:

- Opportunity
- Learner

Each table contains its respective attributes and maintains the integrity of the imported data.

2.4 Verification of Successful Import

The successful import process was verified by:

- Reviewing table previews.
 - Comparing record counts with the source files.
 - Executing SQL queries to confirm accessibility.
 - Inspecting column structures and data types.
-

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Object Explorer

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 - Operators
 - Procedures
 - Sequences
 - Tables (2)
 - learner
 - Columns (256)
 - pk text
 - sk text
 - assigned_coho
 - fee text
 - status text
 - accept_reject_c
 - amount_to_be
 - transaction_id
 - payment_statu
 - i_agree_to_tern
 - modified_at text

Dashboard Properties SQL Statistics Dependencies Dependents Processes postgres/postgres... learner.sql*

postgres/postgres@Superbase

Data Output Messages Notifications

Showing rows: 1 to 20 Page No: 1 of 1

	pk text	sk text	assigned_cohort text	fee text	status text	accept_reject_date text	amount_to_be_paid text	transaction text
1	Learner#	Learner#0000000010Z56C78JF7M6VQXQ2	[null]	0.0	[null]	[null]	[null]	[null]
2	Learner#838ff3db-f3de-4ae4-9c06-29632039ad...	Learner#0000000001R1YRMC22H9RYCY...	[null]	[null]	1010.0	[null]	0.0	[null]
3	Learner#	Learner#0000000010K1EA93ZGXO5SNG	[null]	33.0	[null]	[null]	[null]	[null]
4	Learner#e84aea65-34d3-440a-bf04-f877d3bc5c...	Learner#0000000010MXXW8GCFY55330...	BFSHUCM	[null]	1070.0	1697094086113	0.0	[null]
5	Learner#9f52ded-a970-4700-830a-c13df19749...	Learner#0000000010BMP2H74SMG8HW...	BUSNLYK	[null]	1070.0	1774502025544	0.0	[null]
6	Learner#8d4c12bb-211d-459f-8b2d-653095f9b...	Learner#0000000010V1FQTM6225F4HGBF	[null]	[null]	1030.0	1714217388011	1.0	[null]
7	Learner#6f39a383-aa51-4642-8995-61e49649f...	Learner#0000000010469QFHECKMCD2A...	[null]	[null]	1010.0	[null]	0.0	[null]
8	Learner#c577f426-abe7-4f04-a8db-92cff7ca86bd	Learner#0000000001R1YRMC22H9RYCY...	[null]	[null]	1030.0	1682333589131	200.0	[null]
9	Learner#e84aea65-34d3-440a-bf04-f877d3bc5c...	Learner#0000000010CTBSXC407YQRNB...	B89L7VP	[null]	1070.0	1697088776119	0.0	[null]
10	Learner#6ca14518-2878-4135-adc1-06e6f0605...	Learner#0000000010VM9E97B0CBPMNZ...	B3V0FWZ	[null]	1120.0	1721382458168	0.0	[null]
11	Learner#8e5fadab-9195-4540-81ec-c5473143f8...	Learner#0000000010E1ROT50GSMFEEAF9	BG83LZ2	[null]	1110.0	1720163249696	0.0	[null]
12	Learner#97d3cb92-9f45-48ed-b19e-a486c5194...	Learner#0000000010VM9E97B0CBPMNZ...	B3V0FWZ	[null]	1055.0	1722677607263	0.0	[null]
13	Learner#66616bc8-02aa-4ad8-92b5-b1881b3c8...	Learner#0000000001R1YRMC22H9RYCY...	[null]	[null]	1020.0	1682414444720	200.0	1300
14	Learner#e84aea65-34d3-440a-bf04-f877d3bc5c...	Learner#00000000108S27NZ2J4JRYEAV5	BFSM7YV	[null]	1070.0	1688116609107	0.0	[null]
15	Learner#	Learner#0000000010Z5E2JVQBF6VWW4...	[null]	10.0	[null]	[null]	[null]	[null]
16	Learner#	Learner#0000000010K1H1C6M5K9BC7X...	[null]	0.0	[null]	[null]	[null]	[null]
17	Learner#97d3cb92-9f45-48ed-b19e-a486c5194...	Learner#0000000010TT57SWX523EP32	[null]	[null]	1020.0	1730295521656	100.0	[null]
18	Learner#	Learner#000000001095JKDJB0Y6BP456T	[null]	0.0	[null]	[null]	[null]	[null]
19	Learner#2e5c613a-e7e7-4e04-a794-af9f05a06a...	Learner#00000000109VMYECWD3ZEA...	B1V7AXR	[null]	1050.0	1696233876619	0.0	[null]
20	Learner#a0dda5b3-9c0e-40ee-afc3-4655c1621...	Learner#0000000010VM9E97B0CBPMNZ...	BGYWJKH	[null]	1070.0	1718828058455	0.0	[null]

Total rows: 20 Query complete 00:00:01.610 CRLF Ln 14, Col 2

Fig 2.1 Learner Dataset preview from Postresql using supabase

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Object Explorer

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Dashboard Properties SQL Statistics Dependencies Dependents Processes Opportunity.sql* learner.sql*

postgres/postgres@Superbase

Query Query History

Data Output Messages Notifications

Showing rows: 1 to 20 Page No: 1 of 1

	pk text	opportunity_id text	badge text
1	Opportunit...	Opportunity#000000000GBD0AX3Z6VYRG7...	{\"sk\":\"Badge#000000000G87VEXNKKKTJKE9\",\"ref_properties\":{\"relation_id\":\"K9MSIZ\"}}
2	Opportunit...	Opportunity#000000000GPKZ4N696ZF8V2...	{\"sk\":\"Badge#000000000GAR6V5Z2B7HJPM3\",\"ref_properties\":{\"relation_id\":\"6AJ5FD\"}}
3	Opportunit...	Opportunity#0000000010ZF0Z522FA3Q62...	[null]
4	Opportunit...	Opportunity#000000000G34EPWZAJ6BDP...	{\"sk\":\"Badge#000000000GGQGDZ2TSSHR55G9\",\"ref_properties\":{\"relation_id\":\"76DV7Q\"}}
5	Opportunit...	Opportunity#0000000010ZNX2EXET8MGC9...	[null]
6	Opportunit...	Opportunity#000000000GQHDFWJ9R1HQJ4...	[null]
7	Opportunit...	Opportunity#000000000GQK8HSCCDYA3T...	{\"sk\":\"Badge#000000000GK5A7GFSWAQC3BGV6\",\"ref_properties\":{\"relation_id\":\"SDTUK1\"},{\"sk\":\"Badge#000000001034MKC8KAW\"}}
8	Opportunit...	Opportunity#0000000010ZK125W68Z6BG2E...	{\"sk\":\"Badge#000000001068CP0TFB1Y2D0V0Q\",\"created_at\":\"1727707449137\",\"ref_properties\":{\"relation_id\":\"N2A8ZM\"}}
9	Opportunit...	Opportunity#000000000GRHVKG2HQFQC...	{\"sk\":\"Badge#000000000G1X85MGR4KRAZJKY8\",\"ref_properties\":{\"test\":\"test\",\"relation_id\":\"9GL03J\"}}
10	Opportunit...	Opportunity#0000000010YHJCSGAKECPBF...	[null]
11	Opportunit...	Opportunity#00000000100VYENXHY4WR...	{\"sk\":\"Badge#00000000101XM89PMJ1E3R0WF\",\"created_at\":\"1759841722323\",\"ref_properties\":{\"relation_id\":\"M9VACT\"}}
12	Opportunit...	Opportunity#0000000010ZSD23KTSAQW79...	{\"sk\":\"Badge#0000000010DRCWGM8M4F1JHD4C\",\"created_at\":\"1695366244373\",\"ref_properties\":{\"relation_id\":\"D016IC\"}}
13	Opportunit...	Opportunity#000000001003NQ7TXSH7MH...	{\"sk\":\"Badge#00000000106ZMFG7W91E8DXODS\",\"created_at\":\"1688073184167\",\"ref_properties\":{\"relation_id\":\"YGFMRX\"},{\"sk\":\"Ba\"}}
14	Opportunit...	Opportunity#0000000010ZTJ800YSP4QVM1...	{\"sk\":\"Badge#0000000010VKME8SNX7GD368KG\",\"created_at\":\"1740654428005\",\"ref_properties\":{\"relation_id\":\"FWYKOB\"}}
15	Opportunit...	Opportunity#00000000101NPSAJXGDXAZ7...	{\"sk\":\"Badge#0000000010GBH09YBK3PQ1ZHBV\",\"created_at\":\"1697087432015\",\"ref_properties\":{\"relation_id\":\"SIO22V\"}}
16	Opportunit...	Opportunity#00000000105SP6ESV3G2G9K...	[null]
17	Opportunit...	Opportunity#00000000104N6FHK7NJ98G3...	{\"sk\":\"Badge#00000000109JN4X016JA0Z7KMS\",\"created_at\":\"1721365187137\",\"ref_properties\":{\"relation_id\":\"KFMICC\"}}
18	Opportunit...	Opportunity#0000000010YKH21190PKTC57...	{\"sk\":\"Badge#0000000010XDJHNBFRSHBBQ859\",\"created_at\":\"1744785232344\",\"ref_properties\":{\"relation_id\":\"OGRKKX\"}}

Total rows: 20 Query complete 00:00:01.134 CRLF Ln 10, Col 1

Fig 2.2 Opportunity Dataset preview from PostgreSQL using supabase

CHAPTER 3

DATASET OVERVIEW

3.1 Opportunity Dataset

Include:

- Total number of records- 5733
- Total number of columns- 33
- Primary key- pk
- Important attributes- role, fee, category, location, duration etc
- Data types- text

These columns are the **opportunity** list offered by the **Excelerate**. Each row represents one opportunity (internship, competition, scholarship, job, workshop, etc.)

3.2 Learner Dataset

Include:

- Total number of records- 15029
- Total number of columns- 256
- Primary key- pk, sk
- Important attributes- student status, withdraw students, cohort, payement
- Data types- text

This is **Learner/Application Dataset**, where **each row represents one learner's application to a particular opportunity** (internship, course, scholarship, fellowship, etc.) in Excelerate.

CHAPTER 4

INITIAL SQL ANALYSIS

To understand the datasets before cleaning, several SQL profiling queries were executed.

Opportunity Dataset

The analysis included:

- Total record count
- Total column count
- Duplicate record detection
- Duplicate primary key detection
- Missing value analysis
- Missing percentage calculation
- Data completeness calculation
- Category distribution
- Opportunity location distribution
- Application status distribution
- Payment status distribution
- Cohort distribution
- Fee statistics
- Reward statistics
- Date range analysis
- Monthly trend analysis
- Distinct value identification
- Empty string detection

For every SQL query include:

- SQL Query
- Query Result
- Interpretation

Total Records

Result:

Opportunity Dataset = 5733

Interpretation:

The Opportunity dataset contains 5733 records, indicating sufficient data volume for meaningful exploratory analysis.

Total Columns

Result:

Columns- 33

Interpretation-

These columns are the **opportunity** list offered by the **Excelerate**. Each row represents one opportunity (internship, competition, scholarship, job, workshop, etc.)

Check Duplicate Opportunities

Result:

Duplicate Opportunity_id- 0

Interpretation-

There will be no same two opportunities posted twice, high uniqueness.

Check Missing Data

Result:

Total missing values- 1391

Total missing percentage- 24.27%

Interpretation-

Some columns have the missing percentage high with more than 90 % values are missing , so data is massive.

Overall Completeness

Result:

Total missing values- 4342

Total missing percentage- 71.26%

Interpretation-

Overall completeness is not also good , but the completeness of the columns is the main issue.

Category Distribution

Result:

Total unique categories- 14

Outlier Categories- \"ct\": \"1766057859531\"(1), %22value%22: %221%22 }(1), { %22label%22: %22United Kingdom%22(1), Xploreu(2)

Interpretation-

Total opportunities is available in 14 categories.

Location Distribution

Result:

Total Location available for opportunities- 22

Interpretation-

Lack of standardization in this column.

Duration Distribution

Result:

Highest Duration Opportunity- 99 weeks (G1-1 Opportunity tracking internship)

Outlier- 0 week invalid data, 0 hour, %22value%22: %22Vanuatu%22 }

Interpretation-

Which opportunity is completed in how many days, weeks, minutes is given.

Archived VS Active Opportunities

Result:

Archived-384

Active-141

Null- 5205

Interpretation-

Highest ratio is missing.

Auto Approval Distribution

Result:

Manual- 4917

Auto- 811

Outlier- {"sk":"","PE#1766057859639A","value":"Yemen"},
{"key":"whyDoYouWantToApplyForThisInternship"}

Interpretation-

Most applications are manually approved .

Fee Distribution

Result:

text datatype fee

Highest frequency Fee- 0 fee(free)- 4518, total 106 frequency,

Outlier- {"className":"mb-6", "6.00E+12", "ct":"","1766057859531", {
"label":"Venezuela"}

Interpretation-

Mostly Opportunities are free.

Currency Distribution

Result:

Max Currency- USD(4699), INR(355)

Outlier- { %22label%22: %22Uzbekistan%22, \"""ct\":"1766057859531\\"", %22value%22: %224%22 }

Interpretation-

Most of the payment is completed in US Dollar.

Created Opportunities by month, last date to apply

Cannot be analyzed because the datatype is text , it should be in date time.

Opportunities Without Deadline

Result:

Only 3

Interpretation-

Most of the opportunities have a give deadline to apply.

Learner Dataset

Total Records

Result:

Opportunity Dataset = 15029

Interpretation:

The Opportunity dataset contains 15029 records, indicating sufficient data volume for meaningful exploratory analysis.

Total Columns

Result:

Columns- 256

Interpretation-

Most of the columns are unnamed and only contain the null values this is the data quality issue.

The Learner dataset records every learner application submitted for an opportunity. Each record represents a learner's application and captures the entire application journey.

Check Duplicate Primary Key

Result:

Duplicate primary key(pk) - 304

Interpretation-

304 pk are duplicate , but the duplicate % is low.

Check Missing Data

Result:

Total missing values- 1391

Total missing percentage- 24.27%

Interpretation-

Some columns have the missing percentage high with more than 90 % values are missing , so data is massive.

Overall Completeness

Result:

Total missing values- 4342

Total missing percentage- 71.26%

Interpretation-

Overall completeness is not also good , but the completeness of the columns is the main issue.

Status Distribution

Result:

Null status values- 5978

Outlier- { %22label%22: %22Yemen%22

Interpretation-

Status is important factor.

Payment Status Distribution

Result:

Null- 5995

Unpaid- 8529

Paid- 504

Outlier- { %22label%22: %22Zimbabwe%22

Interpretation-

The unpaid status is more than paid.

Assigned Cohort Distribution

Result:

Null- 9431

Total cohort- 2822

Interpretation-

The section from a particular learner belongs to.

Student Status Distribution

Result:

Null (15023),

prospectiveUndergraduateStudent- 4

prospectiveGraduateStudents-1

Outlier- %22tableView%22: true

Interpretation-

Most of the learner/ student status is missing, only 2 categories are available from which the student belongs.

Fees Distribution, Pending amount-

Cannot analyse because the data type is text first we have to convert into currency.

Withdrawal Analysis

Result:

Total student withdraw- 382

Interpretation-

The 382 out of total withdraw their application.

DropOut Analysis

Result:

Total student dropout- 322

Interpretation-

The 322 out of total dropout their application.

Marketing Source

Result:

Null Values- (14978)

Platforms-

socialMedia, emailFromRochesterInstituteofTechnology,
emailFromSaintLouisUniversity,myFriendRelative, googleSearch,,

Gwendolyn Brooks College Prep, Gwendolyn Brooks College Prep, Kruu Website, Saint
Louis University, other

Outlier- %22type%22: %22select%22, ""size"":""25566""

Interpretation-

There are the source from where the students are engaged about the opportunities provided by
excelerate, so focus on these platforms and other also to increase the applications.

CHAPTER 5

DATA QUALITY ASSESSMENT

5.1 Data Quality Summary

Quality Check	Opportunity Dataset	Learner Dataset
Duplicate Primary Keys	None	None
Duplicate Records	Present	Present
Missing Values	Present	Present
Missing Percentage	24.29%	94.97%
Empty Strings	Few	More
Date Formats	InValid	InValid
Data Types	InCorrect	InCorrect

5.2 Missing Value Analysis

The SQL analysis revealed that several columns contain missing values. Most missing values occur within optional descriptive fields and questionnaire responses rather than mandatory identification fields.

5.3 Duplicate Analysis

Primary key duplication analysis confirmed the uniqueness of records. This indicates that the datasets maintain strong entity integrity.

5.4 Column Naming Consistency

Some columns initially contained inconsistent naming conventions, including uppercase letters and concatenated words. These were standardized to improve SQL readability and maintain consistency throughout the project.

5.5 Date Field Assessment

Date-related attributes were examined for consistency and completeness. Important timeline fields such as creation dates, application dates, completion dates, and deadlines were verified for proper formatting.

5.6 Text Field Assessment

Several long descriptive columns contain detailed textual information intended for users rather than analytical processing. These fields will require preprocessing during later stages if natural language analysis becomes necessary.

CHAPTER 6

INITIAL VISUALIZATIONS

To complement the SQL-based analysis, visualizations were created to summarize important characteristics of the datasets.

Opportunity Dataset Visualizations

- Opportunities by Category
- Opportunities by Location
- Opportunities Created Over Time
- Cohort Distribution
- Fee Distribution
- Active vs Archived Opportunities

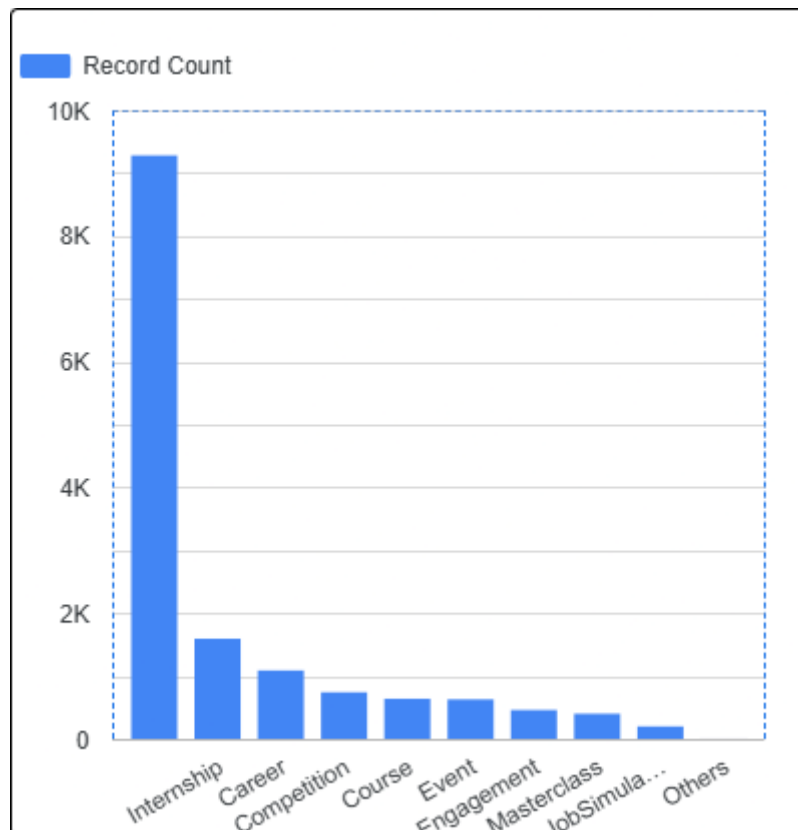


Fig 6.1 Categories Distribution

Interpretation- Most of the learners choose the Internship from the various opportunity provided by the Excelerate.

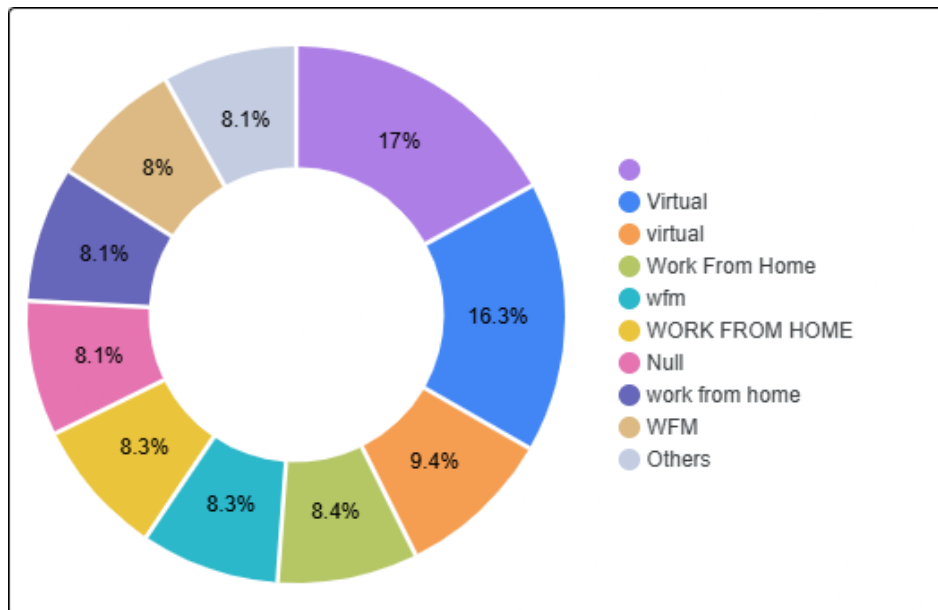


Fig 6.2 Opportunity By Location

Interpretation- There is high standardization issue in location, but most of the opportunities are work from home or virtual.

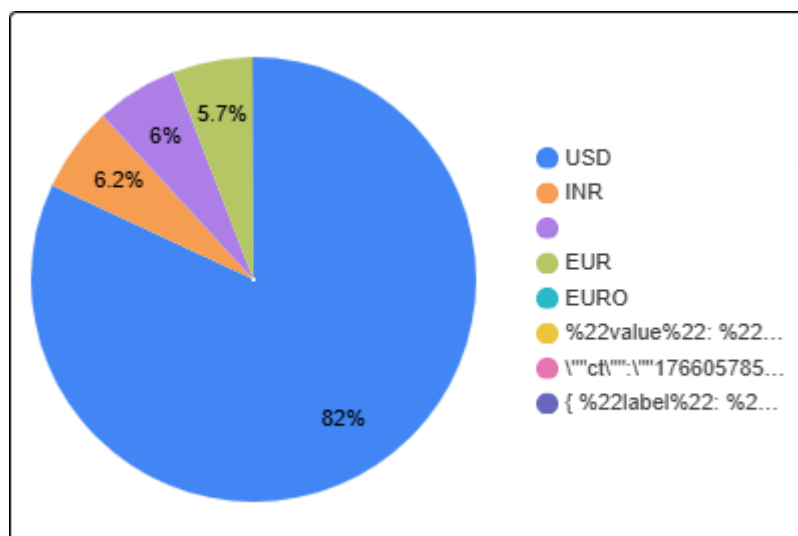


Fig 6.3 Currency Distribution

Interpretation- The most of the payment is done in the US Dollar , outliers are also present(invalid Currency).

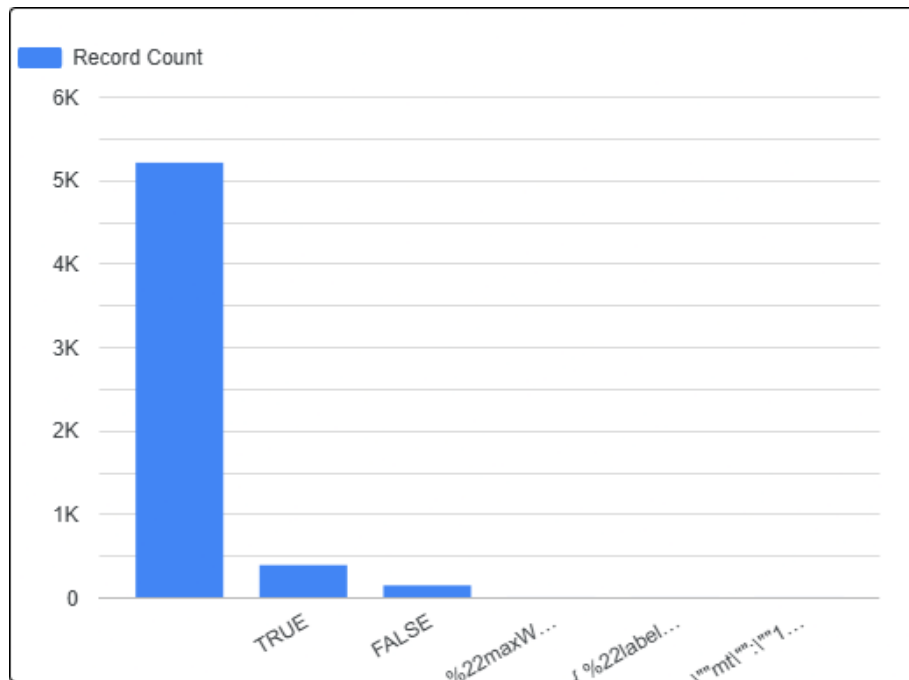


Fig 6.4 Active VS Archived

Interpretation- Here We can find the null values are present mostly but the Archived opportunities are more than active.

Means many of the opportunity are completed are closed their application.

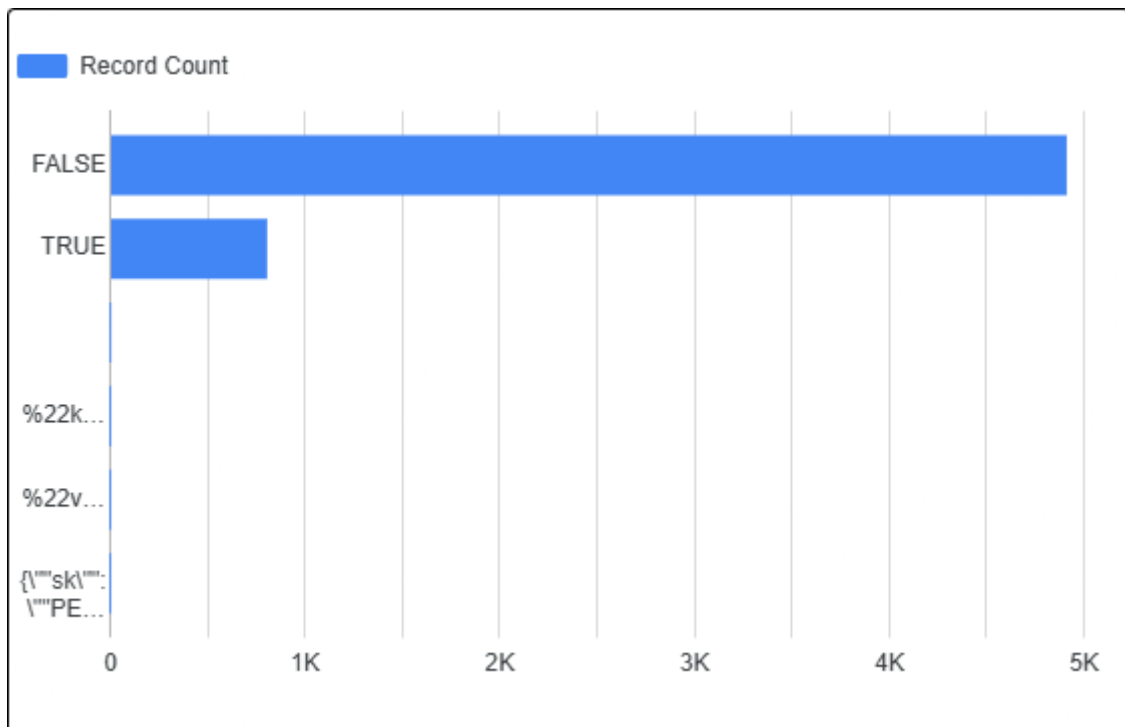


Fig 6.5 Auto Approve Vs Manual Opportunity

Interpretation- The most of the records are approved manually only a small number of applications are auto approved.

Learner Dataset Visualizations

- Application Status Distribution
- Payment Status Distribution
- Student Status Distribution
- Marketing Source Distribution
- Assigned Cohort Distribution
- Applications Over Time

Each visualization provides a high-level understanding of the datasets and assists in identifying trends, imbalances, and potential data quality issues.

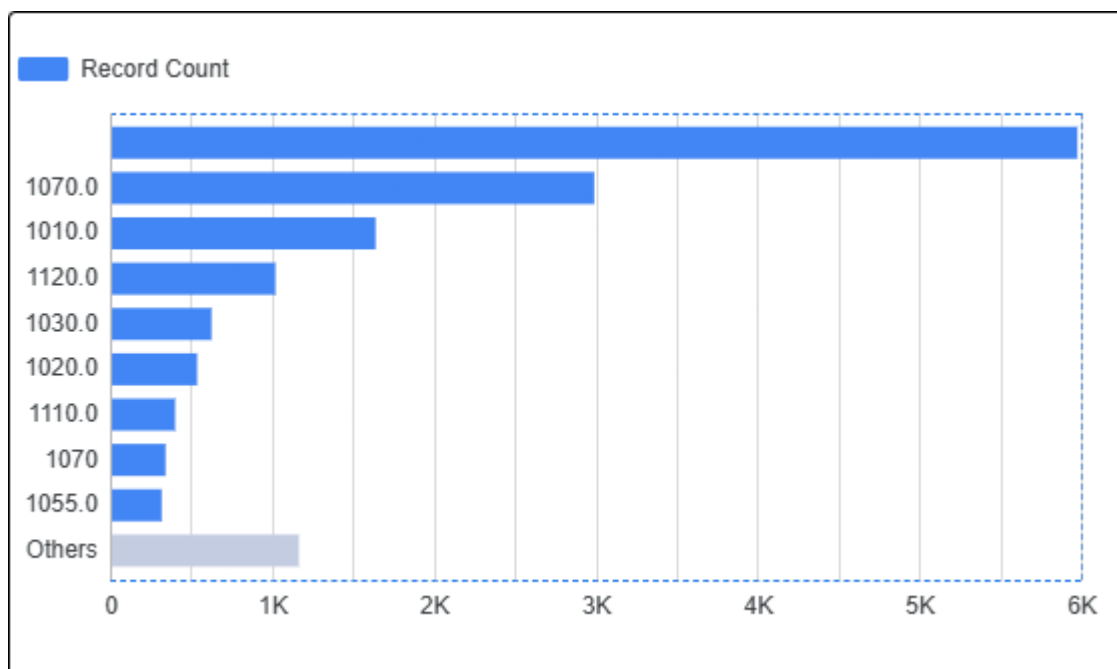


Fig 6.6 Status Distribution

Interpretation- The many learners have the status 1070.0

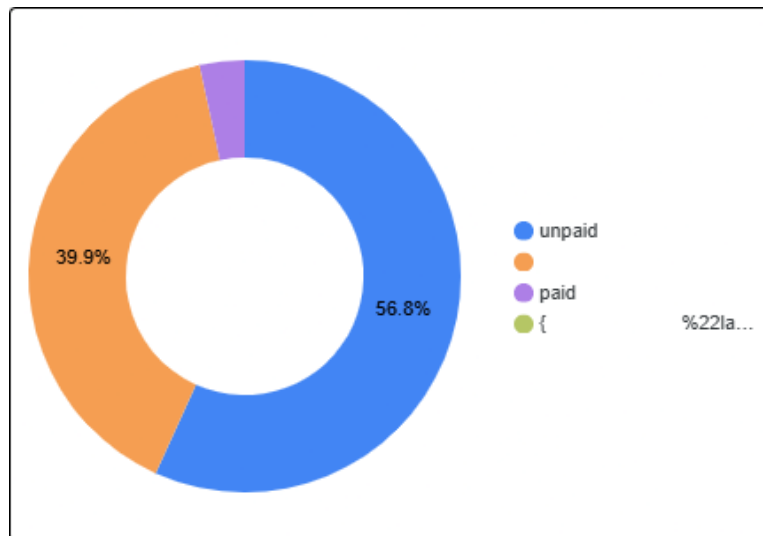


Fig 6.7 Payment Status Distribution

Interpretation- The most of the payments are yet to be paid , only a few student paid the fee.

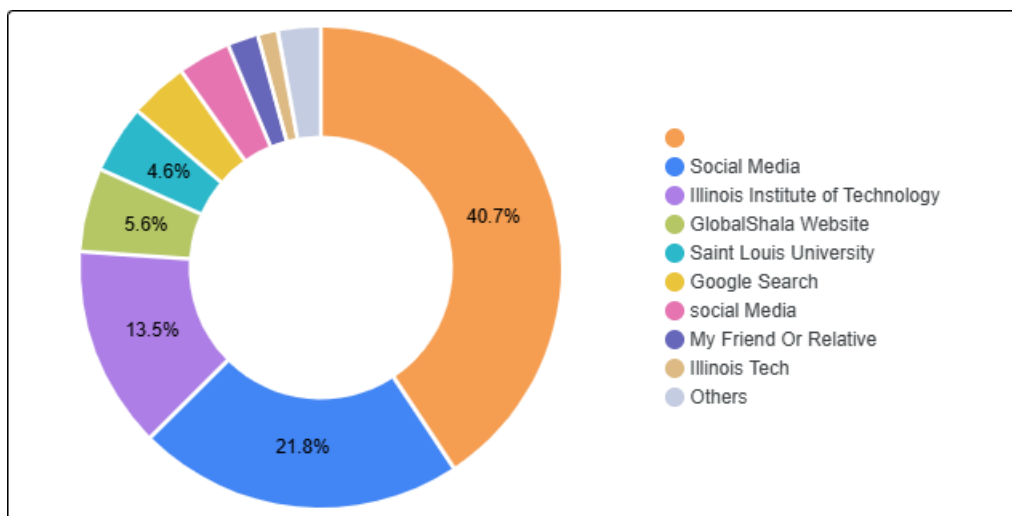


Fig 6.8 Marketing Source

Interpretation- There are only few platforms where the lead is generated.

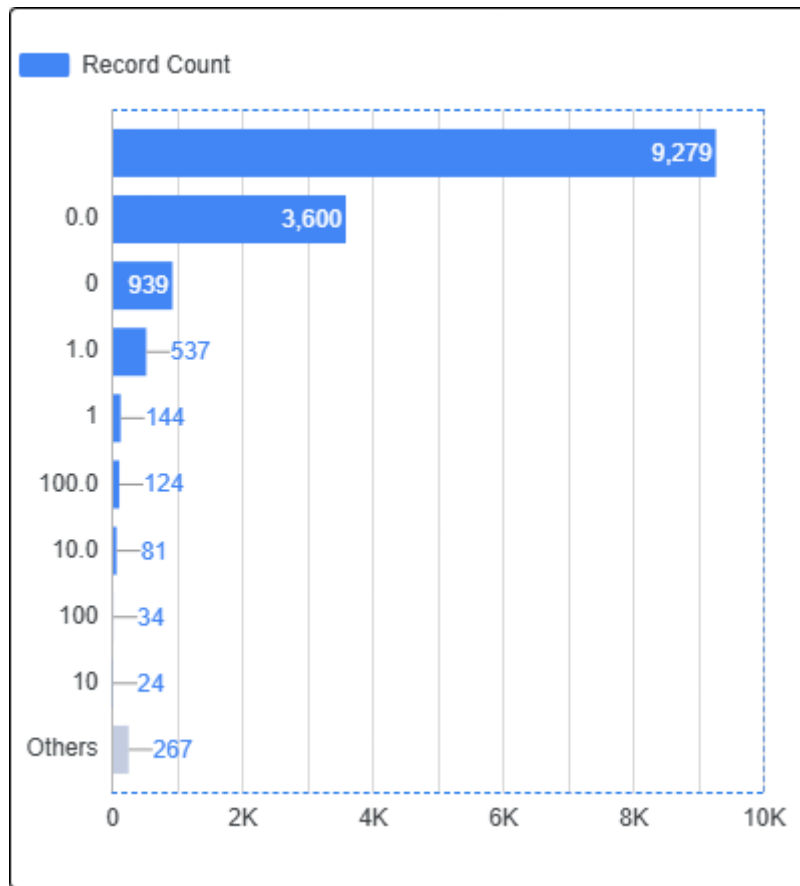


Fig 6.9 Fee Distribution

Interpretation- We can conclude that most of the opportunities are free for learners.

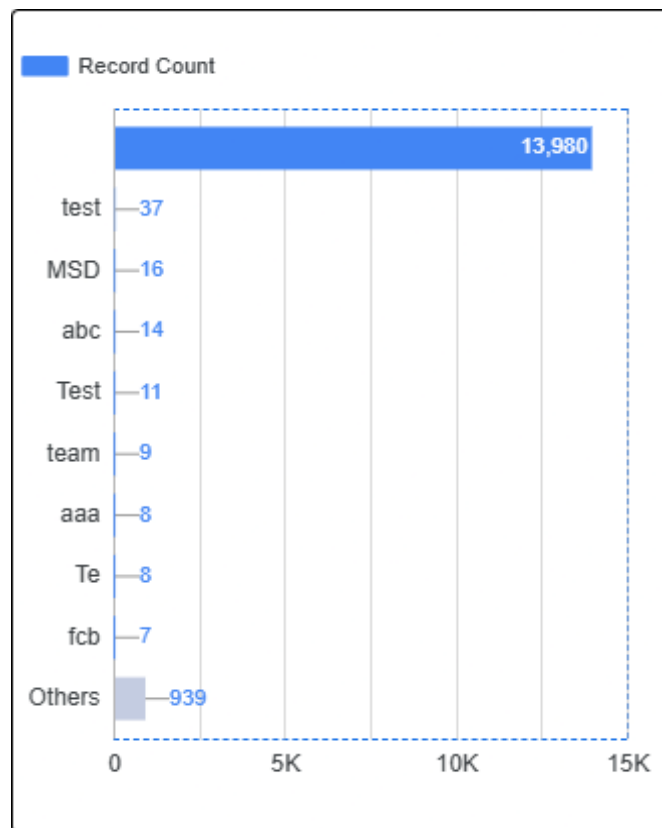


Fig 6.10 Team Applicants

Interpretation- The most candidates lie in the Team Test and there is standardized issue in the particular column.

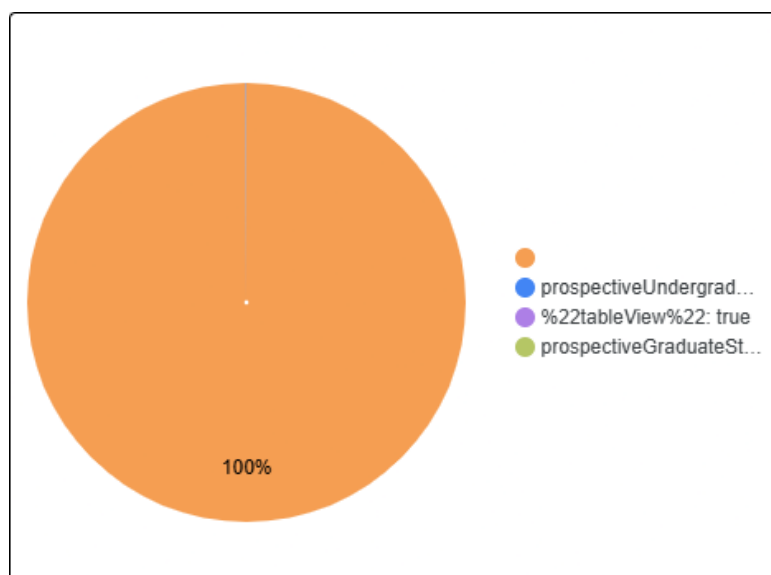


Fig 6.11 Student Status Distribution

Interpretation- Most of the values are missing and only students are related to 2 categories.

CHAPTER 7

OBSERVATIONS

The initial profiling of the datasets resulted in several key observations.

Opportunity Dataset

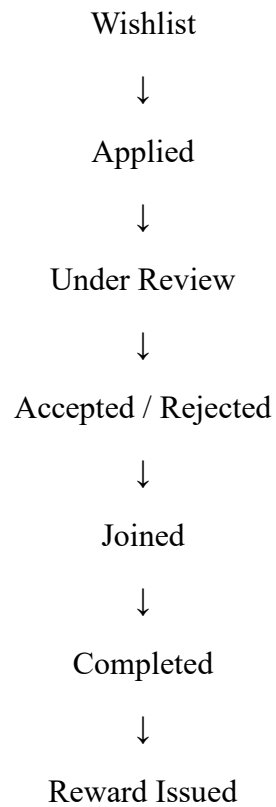
- The dataset contains a diverse range of opportunities across multiple categories.
 - Most opportunities are concentrated within a limited number of categories.
 - Several descriptive fields contain missing values, while primary identification fields are largely complete.
 - Fee-related information indicates a mixture of free and paid opportunities.
 - Date fields appear to be inconsistently formatted.
 - The dataset provides sufficient information for subsequent business analysis.
-

Learner Dataset

- Application records capture the complete learner journey from application through completion or withdrawal.
- Application status values indicate multiple stages of the learner lifecycle.
- Payment-related information enables future financial analysis.
- Questionnaire responses contain a relatively high proportion of missing values, suggesting that some fields are optional.
- Cohort information enables learner segmentation.
- Marketing source information will support future campaign effectiveness analysis.

Application Journey

The lifecycle of a learner can be visualized as:



Overall Assessment

Both datasets are structurally suitable for analysis and demonstrate strong relational potential. While missing values exist in several optional fields, critical identification columns remain largely complete. The datasets are well positioned for the next phase involving data cleaning and transformation.